# SF305

SF305 Halogen-free PI film based flexible copper clad laminate ( 3-layer FCCL)

## FEATURES

- Halogen free, and Flammability UL94 V-0.
- Excellent flexibility, thermal resistance , dimensional stability and electrical properties.
- Compatible with EU RoHS directive, free of Pb, Hg,Cd, $Cr^{6+}$,PBB,PBDE, etc..

## APPLICATIONS

Computer, Mobile phone, Digital camera, VCR, OA equipment, etc..

## GENERAL PROPERTIES

| 性能项目 Test Item | | 试验处理条件 Treatment Condition | 单位 Unit | 性能数据 Property Data | | |
|---|---|---|---|---|---|---|
| | | | | IPC 标准值 Standard | 典型值 Typical Value | |
| | | | | | 051813DR | 051813SE |
| 剥离强度 Peel Strength (90°) | | A | N/mm | ≥0.7 | 1.1 | 1.3 |
| | | 288℃,5s | | ≥0.5 | 1.0 | 1.2 |
| 耐折性 (MIT法) Folding Endurance (MIT) | | R0.8X4.9N | 次 Times | - | >600 | >350 |
| 热应力 Thermal Stress | | 300℃,30s | - | - | 无分层、无起泡 No delamination | 无分层、无起泡 No delamination |
| 尺寸稳定性 Dimensional Stability | MD | E-0.5/150 | % | ±0.2 | ±0.1 | ±0.1 |
| | TD | | | | ±0.1 | ±0.1 |
| 耐化学性（剥离强度保持率） Chemical Resistance | | 暴露化学品后 After Chemical Exposure | % | ≥80 | >85 | >85 |
| 介电常数 (1MHz) Dielectric Constant(1MHz) | | C-24/23/50 | - | ≤4.0 | 3.6 | 3.6 |
| 介质损耗角正切 (1MHz) Dissipation Factor (1MHz) | | C-24/23/50 | - | ≤0.04 | 0.031 | 0.030 |
| 体积电阻率（湿热） Volume Resistivity | | C-96/35/90 | MΩ • cm | $\geq 10^6$ | $1.5x10^8$ | $2.0x10^8$ |
| 表面电阻（湿热） Surface Resistance | | C-96/35/90 | MΩ | $\geq 10^5$ | $5.0x10^6$ | $4.5x10^6$ |

注释 Explanations: C = 湿热处理条件 Humidity conditioning;
E = 恒温处理条件 Temperature conditioning.

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## Product Code Description

SF305 05 18 13 S R

- R-压延铜箔 RA Copper Foil; E-电解铜箔 ED Copper Foil
- S-单面板 Single Side; D-双面板 Double Side
- 胶层厚度 Adhesive Thickness: 13-13μm; 20-20μm
- 铜箔厚度 Copper Foil Thickness: 18-18μm; 35-35μm
- PI膜厚度 PI Film Thickness: 05-12.5μm; 10-25μm
- 生益覆盖膜产品编号 Shengyi Coverlay Designation

## SPECIFICATIONS OF STANDARD PRODUCTS

| 产品规格 Specifications | 材料厚度 Thickness (μm) | | | 单面板/双面板 Single/Double Side | 铜箔类型 Copper Type |
|---|---|---|---|---|---|
| | PI 膜 PI film | 铜箔 Copper Foil | 胶粘剂 Adhesive | | |
| SF305 051213DR | 12.5 | 12 | 13 | 双面板 Double | 压延铜箔 RA |
| SF305 051213DE | 12.5 | 12 | 13 | 双面板 Double | 电解铜箔 ED |
| SF305 051813SR | 12.5 | 18 | 13 | 单面板 Single | 压延铜箔 RA |
| SF305 051813DR | 12.5 | 18 | 13 | 双面板 Double | 压延铜箔 RA |
| SF305 051813SE | 12.5 | 18 | 13 | 单面板 Single | 电解铜箔 ED |
| SF305 051813DE | 12.5 | 18 | 13 | 双面板 Double | 电解铜箔 ED |
| SF305 101820SR | 25 | 18 | 20 | 单面板 Single | 压延铜箔 RA |
| SF305 101820DR | 25 | 18 | 20 | 双面板 Double | 压延铜箔 RA |
| SF305 101820SE | 25 | 18 | 20 | 单面板 Single | 电解铜箔 ED |
| SF305 101820DE | 25 | 18 | 20 | 双面板 Double | 电解铜箔 ED |
| SF305 103520SR | 25 | 35 | 20 | 单面板 Single | 压延铜箔 RA |
| SF305 103520SE | 25 | 35 | 20 | 单面板 Single | 电解铜箔 ED |
| SF305 103520DR | 25 | 35 | 20 | 双面板 Double | 压延铜箔 RA |
| SF305 103520DE | 25 | 35 | 20 | 双面板 Double | 电解铜箔 ED |
| SF305 203520SR | 50 | 35 | 20 | 单面板 Single | 压延铜箔 RA |
| SF305 203520SE | 50 | 35 | 20 | 单面板 Single | 电解铜箔 ED |

## PURCHASING INFORMATION

SF305 is supplied in rolls and standard widths of 250±2mm or 500±2mm. Roll length is 50+1/-0m or 100+2/-0m.Other sizes could be available upon request.

## STORAGE CONDITION

Stored in the room of dryness and no corrosive gases. One year when stored in the original packaging at room temperature. The products do not require refrigeration and should not be frozen.